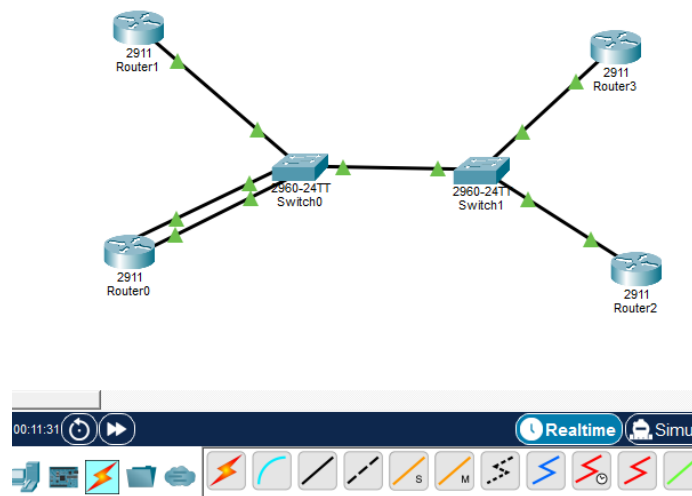


WEEK 2 NETWORKS PRACTICAL

TOPOLOGY



S1 MAC ADDRESS TABLE

```
Switch>enable
Switch#show mac address-table
      Mac Address Table
-----
Vlan    Mac Address      Type    Ports
----    -
1       0002.1636.9018   DYNAMIC Fa0/24
1       0005.5ec6.e801   DYNAMIC Fa0/2
1       00d0.9772.5a01   DYNAMIC Fa0/1
1       00d0.9772.5a02   DYNAMIC Fa0/3
```

S2 MAC ADDRESS TABLE

```
Switch>enable
Switch#show mac address-table
      Mac Address Table
-----
Vlan    Mac Address      Type    Ports
----    -
1       0001.4272.4a01   DYNAMIC Fa0/1
1       0002.172d.3d18   DYNAMIC Fa0/24
1       0060.4745.9e01   DYNAMIC Fa0/2
Switch#
```

IP ADDRESSES

R0 G0/0

IP Configuration	
IPv4 Address	192.168.10.1
Subnet Mask	255.255.255.0

R0 G0/1

IP Configuration	
IPv4 Address	192.168.20.1
Subnet Mask	255.255.255.0

R1

IP Configuration	
IPv4 Address	192.168.10.2
Subnet Mask	255.255.255.0

R2

IP Configuration	
IPv4 Address	192.168.10.3
Subnet Mask	255.255.255.0

R3

IP Configuration	
IPv4 Address	192.168.10.4
Subnet Mask	255.255.255.0

PING RESULTS

```
Router>enable
Router#ping 192.168.10.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.10.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 5/12/33 ms

Router#ping 192.168.10.3

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.10.3, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 11/19/43 ms

Router#ping 192.168.10.4

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.10.4, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 11/19/23 ms
```

Dynamic MAC address table

Switch0

```
Switch>enable
Switch#show mac address-table dynamic
      Mac Address Table
-----
Vlan    Mac Address      Type    Ports
----    -
1       0002.1636.9018   DYNAMIC Fa0/24
1       0005.5ec6.e801   DYNAMIC Fa0/2
1       00d0.9772.5a01   DYNAMIC Fa0/1
1       00d0.9772.5a02   DYNAMIC Fa0/3
```

Switch1

```
Switch>enable
Switch#show mac address-table dynamic
      Mac Address Table
-----
Vlan    Mac Address      Type    Ports
----    -
1       0001.4272.4a01   DYNAMIC Fa0/1
1       0002.172d.3d18   DYNAMIC Fa0/24
1       0060.4745.9e01   DYNAMIC Fa0/2
```

After clearing

```
Switch#clear mac address-table dynamic
Switch#show mac address-table dynamic
      Mac Address Table
-----
Vlan    Mac Address      Type    Ports
----    -
1       0002.1636.9018   DYNAMIC Fa0/24
1       0005.5ec6.e801   DYNAMIC Fa0/2
```

```
Switch#clear mac address-table dynamic
Switch#show mac address-table dynamic
      Mac Address Table
-----
Vlan    Mac Address      Type    Ports
----    -
1       0002.172d.3d18   DYNAMIC Fa0/24
```

IP ROUTE

R0

```
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/24 is directly connected, GigabitEthernet0/0
L       192.168.10.1/32 is directly connected, GigabitEthernet0/0
    192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.20.0/24 is directly connected, GigabitEthernet0/1
L       192.168.20.1/32 is directly connected, GigabitEthernet0/1
```

R1

```
    192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/24 is directly connected, GigabitEthernet0/0
L       192.168.10.2/32 is directly connected, GigabitEthernet0/0
```

R2

```
    192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/24 is directly connected, GigabitEthernet0/0
L       192.168.10.3/32 is directly connected, GigabitEthernet0/0
```

R3

```
    192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/24 is directly connected, GigabitEthernet0/0
L       192.168.10.4/32 is directly connected, GigabitEthernet0/0
```

REFLECTION QUESTION

Yes. MAC addresses reappear because the switch automatically learns the source MAC addresses of devices when they communicate. After the MAC address table is cleared, network traffic causes the switch to learn the MAC addresses again and rebuild its dynamic MAC address table.